

Home Lab Project Week 6, Day 2

Task:

Review the Ubuntu target and perform a basic Nmap SYN scan from Kali Linux. Record the target IP address, discovered ports, and scan results.

Nmap SYN Scan of Ubuntu Target

Engagement: Home Lab Internal Network Assessment **Lab log:** Week 6, Day 1 **Target:** Ubuntu **192.168.78.5** · **Tester:** Kali Linux **192.168.78.3** **Prepared by:** [Mustapha Hirsch], DAE Cybersecurity Program **Date:** 2026-09-23

1. Objective

My objective was to review the Ubuntu target and perform a basic Nmap SYN scan from Kali Linux, then record the target IP address, the discovered ports, and the scan results.

2. What a SYN Scan Is

A SYN scan (**-sS**), also called a "half-open" scan, sends a TCP SYN packet to each port and reads the reply without completing the handshake:

- **SYN-ACK returned** → the port is **open**
- **RST returned** → the port is **closed**
- **No reply** → the port is **filtered** (likely a firewall)

Because it never finishes the three-way handshake, it is fast and quieter than a full-connect scan. It requires raw-packet privileges, so I ran it with **sudo**.

3. Command I Ran

bash

```
sudo nmap -sS -sV -Pn 192.168.78.5
```

- **-sS** — TCP SYN (half-open) scan
 - **-sV** — service/version detection (identifies the software behind each open port)
 - **-Pn** — skip host discovery and treat the host as up (useful if ping is blocked)
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4. Scan Results

Starting Nmap 7.99 (<https://nmap.org>) at 2026-09-23 14:21 -0400

Nmap scan report for 192.168.78.5

Host is up (0.0021s latency).

Not shown: 998 closed tcp ports (reset)

PORT STATE SERVICE VERSION

21/tcp open ftp vsftpd 3.0.5

23/tcp open telnet

MAC Address: 08:00:27:E3:80:BC (Oracle VirtualBox virtual NIC)

Service Info: OS: Unix

Nmap done: 1 IP address (1 host up) scanned in 48.87 seconds

Discovered open ports

Port	State	Service	Version
21/tcp	open	ftp	vsftpd 3.0.5
23/tcp	open	telnet	(version not identified)

5. Observations

- **Host confirmed up.** The low latency (0.0021s) confirms the target is reachable over the host-only network.
- **998 closed ports.** Of the top 1,000 ports scanned, 998 returned RST (closed) — the SYN scan working as intended.
- **FTP (21/tcp) is open**, running **vsftpd 3.0.5**. This is a named, specific version I can look up in the National Vulnerability Database for known issues, and it is a cleartext-credential service like telnet.

- **Telnet (23/tcp) is open.** The version scanner could not fingerprint it because telnet's option-negotiation bytes confused the probe — harmless, and expected for telnet.
 - **MAC 08:00:27:E3:80:BC** identifies the host as an Oracle VirtualBox virtual NIC, confirming I scanned the correct VM.
 - **OS: Unix** — inferred from the service banners, consistent with the Ubuntu target.
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6. Recorded Results

Field	Value
Date	Week 6, Day 1 — 2026-09-23
Attacker (source)	Kali Linux — 192.168.78.3
Target	Ubuntu — 192.168.78.5 (MAC 08:00:27:E3:80:BC)
Command	sudo nmap -sS -sV -Pn 192.168.78.5
Scan type	TCP SYN (half-open), with version detection
Open ports	21/tcp ftp (vsftpd 3.0.5), 23/tcp telnet
Closed ports	998
OS hint	Unix
Scan duration	48.87 seconds

7. Next Step

This scan covered only the top 1,000 ports. To be certain nothing is listening on a higher port, I plan to run a full port scan:

```
bash
sudo nmap -sS -p- 192.168.78.5 -oN week6-day1-fullscan.txt
```